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The Retarded Nonlocal MOND Kernel Is Unstable on MOND Backgrounds: a Longitudinal Gradient Instability and a Deep-MOND Ghost Close the Nonlocal Door

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The Retarded Nonlocal MOND Kernel Is Unstable on MOND Backgrounds: a Longitudinal Gradient Instability and a Deep-MOND Ghost Close the Nonlocal Door

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Abstract

Purely metric nonlocal realizations of MOND (Deffayet, Esposito-Farèse & Woodard 2011; Woodard 2014; Deffayet & Woodard 2026) carry the MOND force in a retarded nonlocal scalar $X = \square_{\text{ret}}^{-1}(R_{\mu\nu}u^\mu u^\nu)$ through an algebraic function $f(Z)$, $Z = (4c^4/a_0^2)\partial_\mu X\partial^\mu X$, and are the one class that escapes the local no-go for a relativistic MOND completion with $\Phi = \Psi$ lensing and $\alpha_3 = 0$. No perturbation analysis of this kernel on a MOND background exists. We perform it. Around a uniform MOND field ($\bar{Z} = 4y^2$, $y = g/a_0$), in Newtonian gauge and plane symmetry, at leading order in $a_0 \rightarrow 0$ at fixed y , the scalar sector of the in-in theory — the auxiliaries eliminated by their retarded particular solutions — has the dispersion $(e + 3)\omega^4 - (2e - 1)k^2\omega^2 + ek^4 = 0$ at $y = 1$ for the exponential law, with discriminant $1 - 16e < 0$: complex frequencies, $\text{Im}\omega = 0.39ck$. The instability is present for every $y \geq 0.5$ tested ($\text{Im}\omega/ck = 0.21, 0.39, 0.44, 0.36, 0.18, 0.05$ at $y = 0.5, 1, 2, 3, 5, 8$), grows without bound in k (e-folding 1.3×10^3 yr at 1 kpc, 13 yr at 10 pc), is longitudinal only (the transverse sector is stable), is driven entirely by $f''(Z)$ — the μ' term without which there is no MOND — and appears identically for Deffayet–Woodard’s own $f(Z) = \frac{1}{2}Ze^{-\sqrt{Z}/3}$. In the real-frequency deep-MOND window ($y \leq 0.25$) one of the two propagating scalar modes has

negative energy. The result holds under both definitions of the theory, the ordinary localized action and the in-in retarded functional, because $\det M = \det D \cdot \det M_{\text{red}}$ and null data removes only the pure-auxiliary poles at $\omega = \pm k$. Every internal check passed first: the GR limit has no scalar metric mode and the clock mode $\omega^2 = u/(1+u)k^2$ exactly; the static longitudinal response is $1/(\mu_{\parallel}k^2)$ with $\mu_{\parallel} = d(y\mu)/dy = 1 - 2f' - 4\bar{Z}f''$; $\Phi = \Psi$; the localized (X, ξ) kinetic block is $[[-4e^{-y}, 1], [1, 0]]$, $\det = -1$. The mechanism is the radial gradient instability that killed the khronometric a_{μ} -coupled class: any dynamical carrier of $\mu(|\nabla\Phi|)$ turns μ' into a wrong-sign longitudinal kinetic term in the transition regime. Consequences: the condensate-clock nonlocal candidate we had built the same night, which passed 29 local-Lagrangian gates, is dead at the stability gate; the Deffayet–Woodard kernel inherits the result; and with the local carriers and the aether already closed, every class examined for the strict completion fails. Version 2 adds two results. The natural repair, projecting the time derivative of X out of Z ($Z_{\perp} = (4c^4/a_0^2)q^{\mu\nu}\partial_{\mu}X\partial_{\nu}X$, statically identical), shifts the unstable window to $y \geq 2$ and keeps the deep-MOND negative-energy mode: the f'' term is spatial and cannot be projected away. And a second, independent kill: on a galaxy background the multiplier of the localized kernel is $\bar{\xi} = 4\Psi_{\text{ph}} \approx -3 \times 10^{-6}$, the tensor dispersion of $\sqrt{-g}[R + \bar{\xi}R_{uu}]$ is $(1 - \bar{\xi})\omega^2 - k^2 = 0$, so $c_T^2 = 1/(1 - \bar{\xi})$ and $|c_T/c - 1| \approx 1.5 \times 10^{-6}$, 2×10^9 above the GW170817 bound — the luminality kill of the curvature-coupled clock class, inherited; every prior “ $c_T = c$ ” certification of this kernel was the linear statement on Minkowski, where $\bar{\xi} = 0$. Scope: linear order, WKB background, plane symmetry, this kernel structure; nonlocal form factors on the Weyl or Einstein tensor are not covered and no action for them exists. All numbers are printed by a committed script with checks that can fail and an ablation control.

1. The door, and what was known

A relativistic completion of MOND is asked to deliver, from one action, exact MOND with the exponential law $\mu(y) = 1 - e^{-y}$, no-slip lensing $\Phi = \Psi$, an acceptable full PPN set including the preferred-frame parameters, matter conservation, luminal gravitons, stability, an expanding cosmology and controlled limits (the 13-gate specification of the companion programme). By 2026-09-01 the local carriers were closed by committed computation: elliptic and algebraic constraint carriers by $\alpha_3 = O(1)$ (an instantaneous constraint cannot be retarded), the frame-free scalar by $O(1)$ slip, the clock-plus-curvature carriers by $c_T \neq c$, the acceleration-coupled khronometric carrier by a radial gradient instability on $a_0 < a < 38a_0$, and the aether by $\alpha_1 = -2(K_B + 2)$. One door remained: a genuinely nonlocal, retarded carrier, for which $\alpha_3 = 0$ because the response $1/(k^2 - \omega^2/c^2)$ carries the retardation.

The concrete instance is the Deffayet–Woodard kernel. With a timelike frame $u_{\mu} = \partial_{\mu}\phi$,

$$X = \square_{\text{ret}}^{-1}(R_{\mu\nu}u^{\mu}u^{\nu}), \quad Z = \frac{4c^4}{a_0^2}g^{\mu\nu}\partial_{\mu}X\partial_{\nu}X, \quad \Delta L = \frac{c^4}{16\pi G}a_0^2f(Z)\sqrt{-g},$$

and in the static weak field $X \rightarrow \Psi$, $Z \rightarrow 4|\nabla\Psi|^2c^4/a_0^2 = 4y^2$, so that the potential obeys $\nabla \cdot [(1 - 2f'(Z))\nabla\Psi] = 4\pi G\rho/c^2$: the MOND function is $\mu = 1 - 2f'(4y^2)$. Deffayet & Woodard use $f = \frac{1}{2}Ze^{-\sqrt{Z}/3}$; the exponential law is obtained exactly by $f_{\text{exp}} = 4 - 2(\sqrt{Z} + 2)e^{-\sqrt{Z}/2}$. The kernel is localized by a multiplier, $\xi(\square X - R_{uu})$, whose ordinary-action Hessian in the (X, ξ) pair is indefinite, $[[-4e^{-y}, 1], [1, 0]]$ with $\det = -1$ (an independent Dirac audit, 2026-09-02): two auxiliary modes, one ghost-signed. The standard resolution is to define the theory in the in-in (Schwinger–Keldysh) sense, with X, ξ as retarded functionals of the metric carrying no Cauchy data. What that theory's

free modes are on a background where the kernel is active had never been computed: Deffayet & Woodard (2026) defer the perturbation analysis; Woodard (2014) lists the ghost question as future work; the certification audit of the kernel in this repository was carried out on Minkowski and FLRW, where $\bar{Z} = 0$ or f is exponentially dead.

We computed it, having first built the strongest candidate the door admits.

2. The candidate, for the record

The condensate-clock nonlocal MOND action (CCNL) replaces Deffayet–Woodard’s mimetic clock and its advected dust by the Aether–Scalar–Tensor condensate clock $K(Q) = -2\Lambda + K_2(Q - Q_0)^2$, $Q = \sqrt{-(\partial\phi)^2}$, $u_\mu = \partial_\mu\phi/Q$, whose dust is a $\gamma = 2$ polytrope with $c_s^2 = |\Psi|c^2$ inside wells and therefore galaxy-safe (Zenodo 22242701), and uses f_{exp} . It passed 29 committed gates: the galaxy radial-acceleration relation shifts by ≤ 0.001 dex at the full cosmic dust share (Deffayet–Woodard’s pressureless dust shifts it by 0.06–0.24 dex), the off-shell Noether identity of the localized action holds to machine precision with the retarded kernel (so the “hand-imposed retarded operator is not Euler–Lagrange” objection is void: retardation is a choice of solution), $R_{uu}^{(1)}[h^{TT}] = 0$ so $c_T = c$, $\gamma - 1 = -e^{-y}$ with $y = 10^{12}$ at the Cassini radius, the clock’s preferred-frame back-reaction is below 10^{-25} , $\alpha_3 = 0$ by structure, the clock is healthy on its positive branch, and a nonlinear minisuperspace integration keeps the auxiliaries slaved and the constraint conserved to 10^{-10} . None of that survives §4.

3. Setup and internal checks

Newtonian gauge, $ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)d\mathbf{x}^2$, fields of (t, x) , $c = 1$, $16\pi G = 1$. Background: $\bar{X} = sx$ with $s = ya_0$ (so $\bar{Z} = 4y^2$), $\bar{\xi} = 0$ (a uniform gradient sources no ξ), the clock at rest with charge u_b ($\bar{Q} = Q_0(1 + u_b)$, $c_s^2 = u_b/(1 + u_b)$). The quadratic action is built from the covariant pieces: $\sqrt{-g}R$ to second order; $\sqrt{-g}a_0^2f(Z)$ expanded around $\bar{Z}$ keeping every term carrying s ; $\sqrt{-g}\xi(\Box X - R_{uu})$ at second order; $\sqrt{-g}K(Q)$ with an exact-to-second-order expansion of Q . Linear Euler–Lagrange equations, plane waves $e^{i(kx - \omega t)}$, and the 5×5 matrix $M(\omega, k)$ on $(\Phi, \Psi, \delta X, \delta\xi, \delta\phi)$. Leading order in $a_0 \rightarrow 0$ at fixed y removes the vacuum-energy and post-Newtonian terms (all background residuals are $O(a_0^2)$: Λ , $a_0^2f_0$, $f's^2$, $K_2Q_0^2u_b$) and leaves the $O(1)$ MOND structure through $f'(\bar{Z})$, $f''(\bar{Z})$. In-in reduction: the auxiliary rows are eliminated by their particular solutions, the Schur complement $M_{\text{red}} = A - BD^{-1}C$; $\det D = -(k - \omega)^2(k + \omega)^2$ are the pure-auxiliary poles that null data removes.

internal check	result
GR + clock, kernel off	(Φ, Ψ) block $\det = -16k^4$: no scalar metric mode; clock $\omega^2 = \frac{1}{21}k^2 = \frac{u_b}{1+u_b}k^2$ exactly ($u_b = \frac{1}{20}$)
ordinary local auxiliary block, $y = 1$	$[[-4e^{-1}, 1], [1, 0]]$, $\det = -1$: the Dirac audit’s matrix, reproduced independently
static longitudinal response, $y = 1$	$\Psi k^2/\rho = -\frac{1}{4} = -1/(4\mu_{\parallel})$ with $\mu_{\parallel} = 1 - 2f' - 4\bar{Z}f'' = d(y\mu)/dy = 1.000$; $\Phi = \Psi$

4. The dispersion

At $y = 1$ with f_{exp} the reduced (Φ, Ψ) determinant factors to

$$-\frac{256 k^4 (k^2 - 21\omega^2) [(e+3)\omega^4 - (2e-1)k^2\omega^2 + e k^4] e^{-1}}{105 (k-\omega)^2 (k+\omega)^2},$$

the clock factor times a quartic whose discriminant is $(2e-1)^2 - 4e(e+3) = 1 - 16e < 0$. Its roots are $\omega^2/k^2 = 0.388 \pm 0.570i$, i.e. $\omega/k = 0.734 \pm 0.388i$: a growing and a decaying pair, growth $\propto k$.

test	outcome
y -scan, f_{exp}	real ω^2 at $y = 0.10$ (0.056, 0.897) and 0.25 (0.174, 0.717); complex at $y = 0.5, 1, 2, 3, 5, 8$ with $\text{Im } \omega/k = 0.21, 0.39, 0.44, 0.36, 0.18, 0.05$
Deffayet–Woodard’s $f = \frac{1}{2}Z e^{-\sqrt{Z}/3}$	complex at $y = 0.5, 1, 2, 4$ ($\text{Im } \omega/k$ up to 0.48); real at 0.25 and 8
ablation $f'' \rightarrow 0$ at fixed f'	stable at $y = 1, 2$ (one real mode, $\omega^2/k^2 = 0.30, 0.61$): the instability is the f'' term
ablation $f' \rightarrow 0$ at fixed f''	unstable ($\text{Im } \omega/k = 0.60, 0.52$)
transverse propagation, $\mathbf{k} \perp \nabla \bar{X}$	stable at $y = 0.5, 1, 2, 4$ ($\omega^2/k^2 = 0.14, 0.30, 0.61, 0.93$): longitudinal only
energy sign, deep-MOND window	$y = 0.10$: $\omega/k = 0.236$ ($E > 0$) and 0.947 ($E = -7.8 \times 10^3$); $y = 0.25$: 0.417 (+), 0.847 (-4.1×10^2)
metric content of the unstable mode, $y = 1$	$ \Phi ^2 + \Psi ^2$ share 6.4%, (X, ξ) share 86%: nonzero, hence a mode of the reduced metric equation
e-folding times, $y = 1$	1 kpc: 1.3×10^3 yr; 10 pc: 13 yr; 1000 AU: 2.4 days
e-folding times, $y = 8$	1 kpc: 10^4 yr; 10 pc: 100 yr; 1000 AU: 18 days

Because $\det M = \det D \cdot \det M_{\text{red}}$, the quartic’s roots are modes of the ordinary local action as well; the in-in prescription removes only $\det D = 0$, the pure-auxiliary modes at $\omega = \pm k$. The unstable modes carry metric content and survive under either definition.

5. Mechanism

MOND requires $\mu' \neq 0$. The static AQUAL operator is elliptic and safe, with longitudinal coefficient $\mu_{\parallel} = \mu + y\mu' > 0$ for the exponential law, and the kernel reproduces exactly that static response (§3). Its time-dependent completion is not safe: the quadratic kernel Lagrangian contributes $-4f_1(\partial_t \delta X)^2 + 4(f_1 + 2\bar{Z}f_2)(\partial_x \delta X)^2$, a wrong-sign time-kinetic term for X in the localized pair, and the f_2 (curvature, μ') part of the longitudinal gradient term is what turns the coupled (Φ, Ψ, X, ξ) dispersion complex once $\bar{Z}$ is finite. With $f'' = 0$ — f linear in Z , μ constant, no MOND — the sector is stable. This is the mechanism that killed the khronometric a_{μ} -coupled class on $a_0 < a < 38a_0$ (2026-08-31): a dynamical carrier of $\mu(|\nabla \Phi|)$ converts μ' into a longitudinal instability in the transition regime. Retardation does not cure it; it only relocates the carrier.

5a. The projected kernel: the natural repair, tested

The mechanism of §5 points at the term $-4f'(Z)(\partial_t X)^2$. AeST’s MOND invariant avoids such a term by projection onto the frame’s spatial slice, and this kernel already carries a frame, so the same projection is free: $Z_\perp = (4c^4/a_0^2) q^{\mu\nu} \partial_\mu X \partial_\nu X$, $q^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. Statically $Z_\perp = Z$: the MOND law, lensing, PPN screening and every static gate are unchanged. The same computation as §3–4 with $Z \rightarrow Z_\perp$ (`ccnl_projected_kernel_linear_2026.py`, 13 checks; GR limit, $\mu_\parallel$, $\Phi = \Psi$ and the Dirac block reproduced) gives:

test	projected kernel
f_{exp} , y -scan	real ω^2 at $y \leq 1$ ($y = 1$: no propagating mode, $\mu_\parallel = 1$ exactly there); complex at $y = 2, 3, 5, 8$ with $\text{Im } \omega/k = 0.37, 0.31, 0.16, 0.05$
Deffayet–Woodard’s f	complex at $y = 1, 2, 4$ ($\text{Im } \omega/k$ up to 0.45); real at 0.25, 0.5, 8
ablation $f'' \rightarrow 0$	stable (control)
transverse propagation	stable at $y = 0.5, 1, 2, 4$
energy sign, real-frequency y	one negative-energy mode at $y = 0.10, 0.25, 0.5$ ($E = -188, -176, -166$)

The projection removes the $y = 0.5$ –1 instability and nothing else: the f'' (μ') term is spatial, it still couples through the indefinite localization pair, and the window shifts up while the ghost stays. This door closes at the same order.

5b. A second, independent kill: $c_T \neq c$ on galaxy backgrounds

Every “ $c_T = c$ ” statement for this kernel class — Deffayet & Woodard’s, the certification audit’s, and §2’s E1–E2 — is the linear identity $R_{uu}^{(1)}[h^{TT}] = 0$ evaluated on a background where the multiplier vanishes. On a galaxy it does not vanish. The static multiplier equation of the localized action is $\nabla^2 \bar{\xi} = 8 \nabla \cdot (f' \nabla \bar{X}) = 4 \nabla^2 \Psi_{\text{ph}}$ (with $16\pi G = 1$), so $\bar{\xi} = 4\Psi_{\text{ph}}$: it is sourced by the MOND phantom density and is of order v^2/c^2 in every MOND zone. A background $\bar{\xi}$ multiplies R_{uu} at second order in the graviton, and $R_{uu}^{(2)}[h^{TT}] = h\ddot{h} + \frac{1}{2}\dot{h}^2 \neq 0$. The TT quadratic action of $\sqrt{-g}[R + \bar{\xi}R_{uu}]$ gives the dispersion

$$(1 - \bar{\xi})\omega^2 - k^2 = 0, \quad c_T^2 = \frac{1}{1 - \bar{\xi}},$$

which is the curvature-coupled clock’s $c_T^2 = 1/(1 - 2\lambda)$ with $\lambda = \bar{\xi}/2$; $\bar{\xi} = 0$ restores $c_T = 1$ (control). On a Milky-Way-like galaxy with the exponential law, $\Psi_{\text{ph}}/c^2 = -7.3 \times 10^{-7}$ at 8 kpc, so $\bar{\xi} = -2.9 \times 10^{-6}$ (8 kpc), -2.0×10^{-6} (30 kpc), and $|c_T/c - 1| \approx |\bar{\xi}|/2 = 1.5 \times 10^{-6}$ against GW170817’s 7×10^{-16} : an excess of 2×10^9 (`ccnl_kernel_tensor_speed_2026.py`, 6 checks). A path average over the MOND zone’s share of the GW170817 path reduces this by 10^{-1} – 10^{-2} , not by 10^9 . Once localized on a galaxy, the retarded kernel is a clock coupled to R_{nn} — the class the programme had already excluded by luminosity — and it inherits that exclusion independently of §4. In the unlocalized in-in theory the same term arises from the second-order expansion of $f(Z[\square^{-1}R_{uu}])$, so this is not an artifact of localization.

6. Consequences

1. **CCNL is dead at the stability gate**, its 29 other passes notwithstanding.
2. **The Deffayet–Woodard kernel inherits the result.** Their 2026 model retains the same X , Z , and an algebraic f ; the difference in the clock sector does not enter at this order (the clock decouples from the kernel at linear order around a vacuum background, $\bar{R}_{\mu\nu} = 0$).
3. **The nonlocal door is closed at linear-WKB order, twice.** The kernel fails at two independent gates, stability (§4, §5a) and tensor speed (§5b), for either interpolation function, either time-projection of Z , and either definition of the theory. With the local constraint carriers (α_3), the frame-free scalar (slip), the curvature-coupled clock (c_T), the acceleration-coupled clock (radial instability) and the aether (α_1) closed by committed computation, every class examined for the strict completion fails. In the companion programme’s language this is Outcome B, a no-go, not Outcome A.
4. **What survives is unchanged:** MOND phenomenology with the acceleration scale $a_0 = c^2 \sqrt{\Lambda/32\pi}$ as input, the falsifiable $a_0 \propto \sqrt{\rho_{\text{DE}}(z)}$, and the condensate dark field of the AeST class, whose cluster yield is 23–33% of the core residual (Zenodo 22242701). No relativistic completion in hand passes the stability gate.

7. Scope, stated honestly

Linear order. A uniform-gradient WKB background, $a_0 \rightarrow 0$ at fixed y ; since the growth is $\propto k$, the short-wavelength regime where WKB is best is the dangerous one. Plane symmetry and Newtonian gauge. The kernel structure: u -projected $R_{\mu\nu}$, algebraic $f(Z)$, no higher-derivative terms; Deffayet & Woodard state their philosophy is an IR modification, and a UV completion with higher spatial derivatives capping the growth at some k_{max} is not part of any published action. Not covered: nonlocal form factors acting on the Weyl or Einstein tensor (a “field-dependent spin-2” completion, for which no action exists in the literature or in this repository) — though a frame-free one is caught by the slip-lock theorem, which needs a frame for MOND lensing, and a frame-coupled one is the class killed here; the fully nonlinear problem; bounded backgrounds with their own scale. The negative-energy mode in deep MOND is a linear result on the same background.

8. Attribution

The kernel is Deffayet, Esposito-Farèse & Woodard (2011), Woodard (2014) and Deffayet & Woodard (2026). The Dirac structure of the localized pair for this candidate is the independent 2026-09-02 audit in the repository (it also caught a placeholder check in the candidate’s gate script, since replaced by a derived one). The radial-gradient mechanism for μ' -carrying relativistic MOND is the repository’s khronometric result of 2026-08-31. The perturbation analysis of the retarded kernel on a MOND background, the in-in reduction with its internal checks, the f'' ablation and the transverse contrast are new here as far as searched; no such analysis is known to us in the literature.

9. Two-sided summary

For the nonlocal programme: the retarded kernel genuinely does what the local carriers cannot — $\alpha_3 = 0$, $\Phi = \Psi$ at 1PN, exponentially screened PPN, luminal gravitons, conservation as a Noether identity of the localized action, and, with a condensate clock, galaxy-safe dust. Those are real and were computed.

Against: on any background where the kernel is active, the scalar sector is unstable in the transition regime with growth times of years at parsec scales and unbounded in k , and ghostly in deep MOND; the instability is the MOND function's own curvature μ' and cannot be removed without removing MOND; Deffayet–Woodard's own interpolation function shows it. The last door is shut at this order.

10. Reproducibility

- In the directory `qwen_claude_field_theory/ closure_2026/ candidate_ccnl_2026/`: `ccnl_inin_linear_scalar_2026.py` (12 checks, rc=0; output `.out`): §3–5. The physics checks are findings that can fail to materialize; the $f'' \rightarrow 0$ ablation is the control.
- `ccnl_mond_gates_2026.py` (29 checks, rc=0; MUTATE=1 breaks the galaxy comparison): §2.
- `ccnl_projected_kernel_linear_2026.py` (13 checks, rc=0): §5a.
- `ccnl_kernel_tensor_speed_2026.py` (6 checks, rc=0; control $\xi \rightarrow 0$): §5b.
- `ccnl_action_dirac_audit_2026.py` with its tests and manifest (the independent audit): the (X, ξ) block.
- `CCNL_MOND_CANDIDATE.md`, `NONLOCAL_KERNEL_LINEAR_INSTABILITY_VERDICT.md`: the gate table and the verdict.

Quarantine: a_0 (both footings, 9.36×10^{-11} and 1.13×10^{-10} m s $^{-2}$), κ and the clock's charge are inputs. Nothing here derives them.

11. References

- Deffayet C., Esposito-Farèse G., Woodard R. P., 2011, PRD 84, 124054 (arXiv:1106.4984)
- Deffayet C., Esposito-Farèse G., Woodard R. P., 2014, PRD 90, 064038 (arXiv:1405.0393)
- Deffayet C., Woodard R. P., 2026, JCAP 04, 081 (arXiv:2512.10513)
- Deser S., Woodard R. P., 2013, JCAP 11, 036 (arXiv:1307.6639)
- Skordis C., Złośnik T., 2021, PRL 127, 161302 (arXiv:2007.00082)
- Woodard R. P., 2014, Can. J. Phys. 93, 242 (arXiv:1403.6763)
- Zimmerman C. P., 2026, Zenodo 10.5281/zenodo.20779562 (the cluster no-go); 10.5281/zenodo.22242701 (the cluster polytrope)